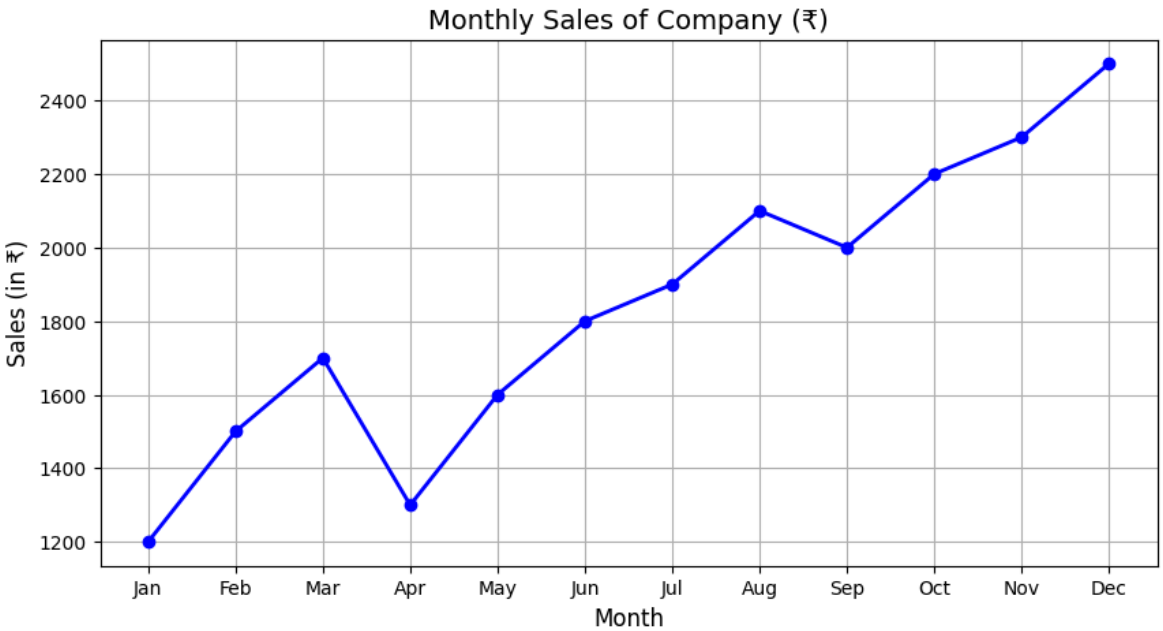
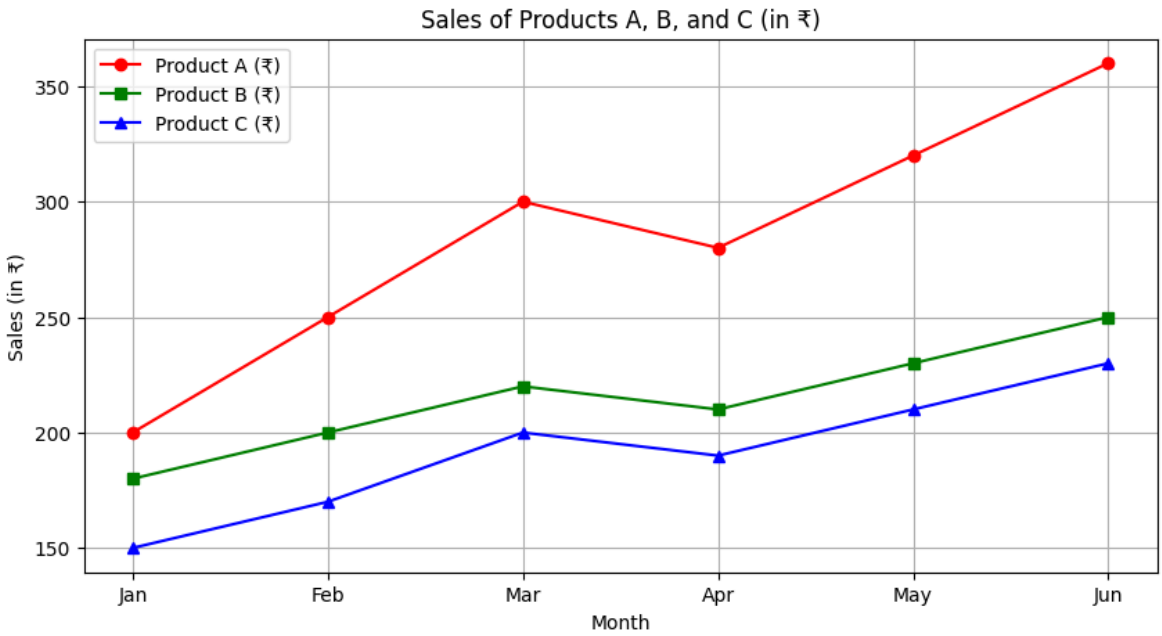
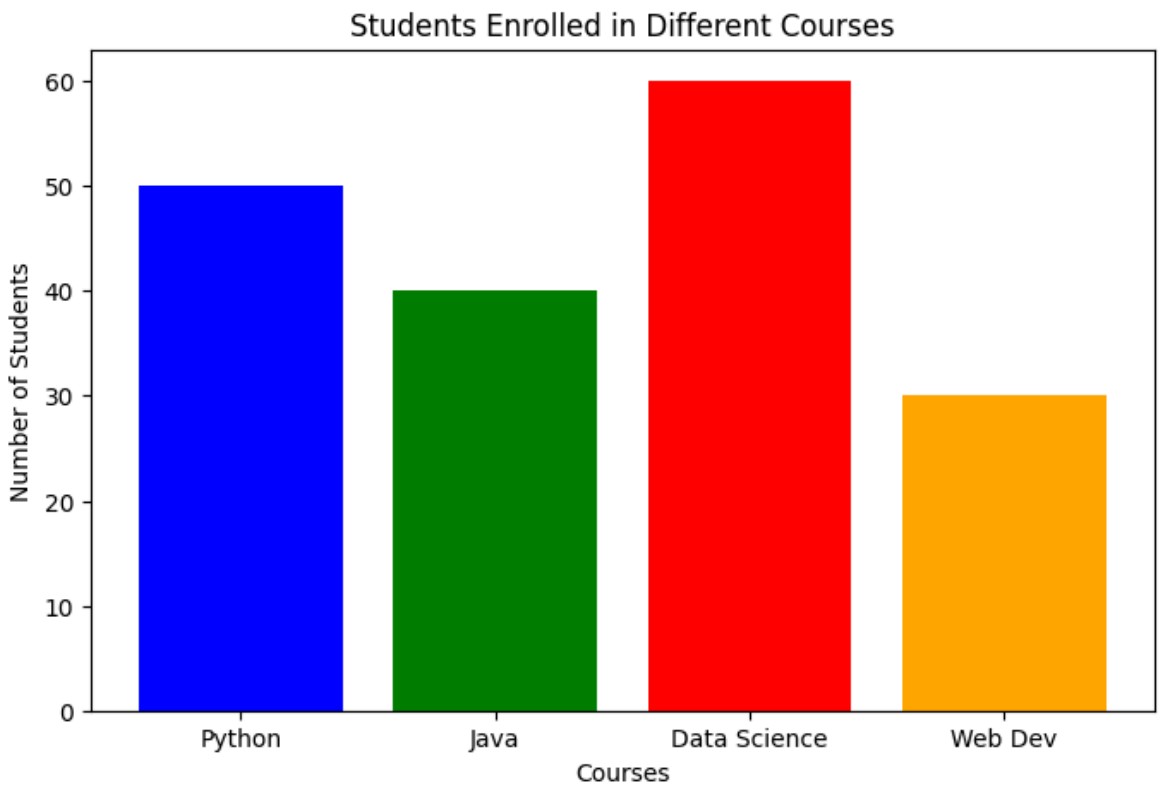
In [1]:
```
import matplotlib.pyplot as plt

# ------------------------------
# Data Preparation
# ------------------------------

# Months of the year
months = ['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','Nov','Dec

# Monthly sales data (in Rupees ₹)
monthly_sales = [1200, 1500, 1700, 1300, 1600, 1800, 1900, 2100, 2000, 2200, 230

# ------------------------------
# Plot Creation
# ------------------------------

# Create figure with size (width=10, height=5)
plt.figure(figsize=(10,5))

# Plot line graph
plt.plot(
    months,
    monthly_sales,
    color='blue',          # Line color
    marker='o',            # Circle markers on each point
    linestyle='-',         # Solid line
    linewidth=2            # Thickness of line
)

# ------------------------------
# Labels and Title
# ------------------------------

plt.title('Monthly Sales of Company (₹)', fontsize=14)   # Title with Rupees sym
plt.xlabel('Month', fontsize=12)                        # X-axis label
plt.ylabel('Sales (in ₹)', fontsize=12)                 # Y-axis label with unit

# ------------------------------
# Grid for better readability
# ------------------------------

plt.grid(True)

# ------------------------------
# Show the graph
# ------------------------------

plt.show()
```



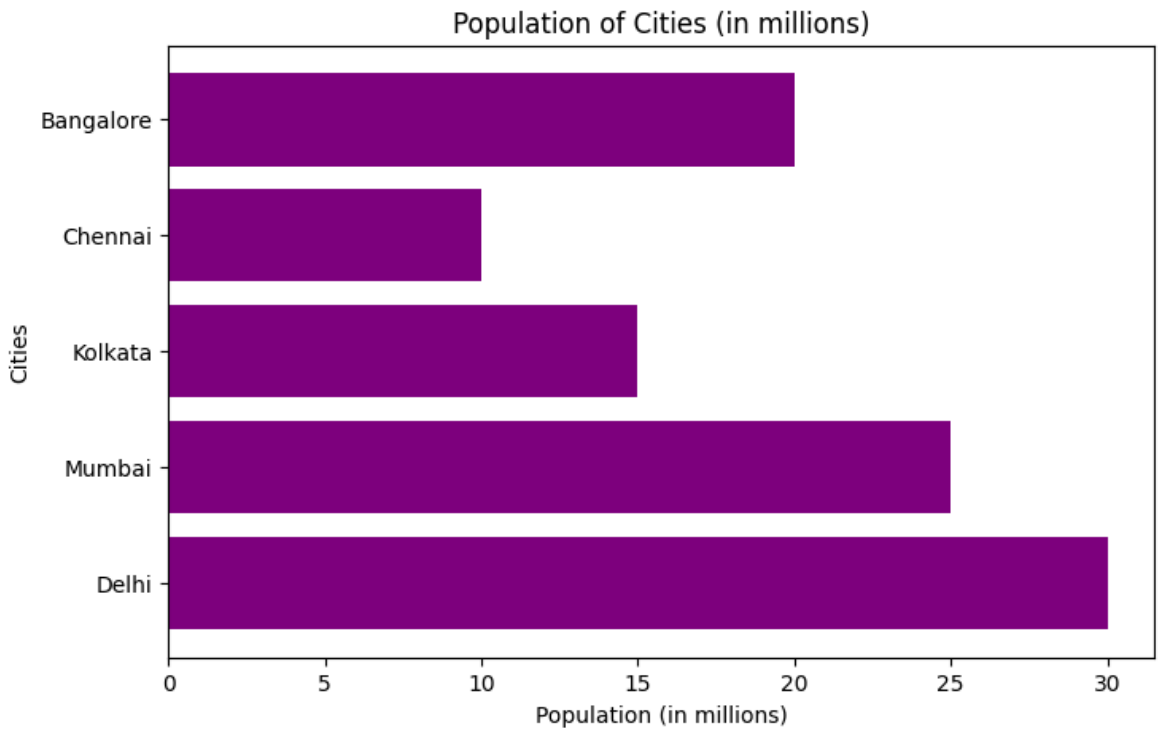
In [2]:
```
import matplotlib.pyplot as plt

# Months for first 6 months
months_6 = ['Jan','Feb','Mar','Apr','May','Jun']

# Sales data of products (in Rupees ₹)
product_a = [200, 250, 300, 280, 320, 360]
product_b = [180, 200, 220, 210, 230, 250]
product_c = [150, 170, 200, 190, 210, 230]

# Create figure
plt.figure(figsize=(10, 5))

# Plot for Product A
plt.plot(months_6, product_a, marker='o', color='red', label='Product A (₹)')

# Plot for Product B
plt.plot(months_6, product_b, marker='s', color='green', label='Product B (₹)')

# Plot for Product C
plt.plot(months_6, product_c, marker='^', color='blue', label='Product C (₹)')

# Title and labels with units
plt.title('Sales of Products A, B, and C (in ₹)')
plt.xlabel('Month')
plt.ylabel('Sales (in ₹)')

# Show legend to identify each product
plt.legend()

# Add grid for clarity
plt.grid(True)

# Display graph
plt.show()
```



In [3]:
```
import matplotlib.pyplot as plt

# Course names
courses = ['Python', 'Java', 'Data Science', 'Web Dev']

# Number of students enrolled in each course
students_enrolled = [50, 40, 60, 30]

# Create figure
plt.figure(figsize=(8, 5))

# Create bar chart
plt.bar(
    courses,
    students_enrolled,
    color=['blue', 'green', 'red', 'orange']  # Different colors for each bar
)

# Title and axis labels
plt.title('Students Enrolled in Different Courses')
plt.xlabel('Courses')
plt.ylabel('Number of Students')

# Display the chart
plt.show()
```

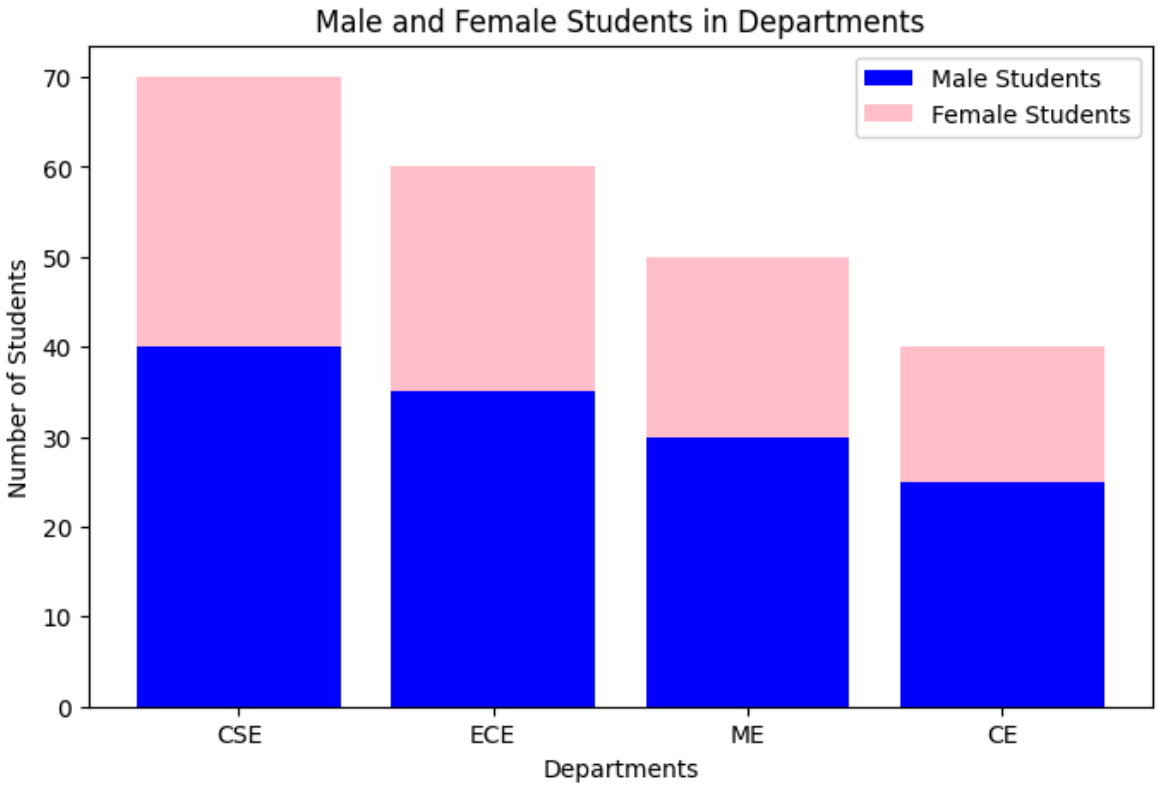


In [4]:
```
import matplotlib.pyplot as plt

# List of cities
cities = ['Delhi', 'Mumbai', 'Kolkata', 'Chennai', 'Bangalore']

# Population data (in millions)
population = [30, 25, 15, 10, 20]

# Create figure
plt.figure(figsize=(8, 5))

# Create horizontal bar chart
plt.barh(
    cities,
    population,
    color='purple'   # Bar color
)

# Title and axis labels with units
plt.title('Population of Cities (in millions)')
plt.xlabel('Population (in millions)')
plt.ylabel('Cities')

# Display the chart
plt.show()
```

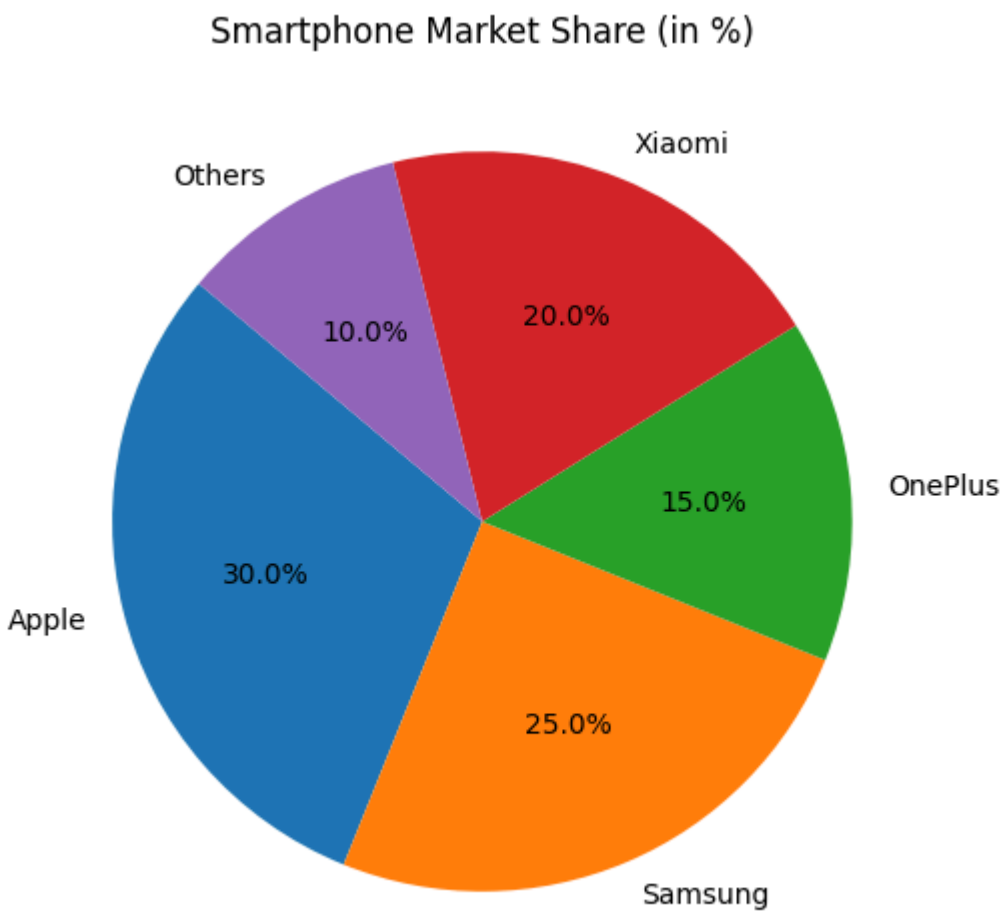


In [5]:
```
import matplotlib.pyplot as plt

# Departments
departments = ['CSE', 'ECE', 'ME', 'CE']

# Number of male and female students
male_students = [40, 35, 30, 25]
female_students = [30, 25, 20, 15]

# Create figure
plt.figure(figsize=(8, 5))

# Bar chart for male students
plt.bar(
    departments,
    male_students,
    color='blue',
    label='Male Students'
)

# Stacked bar for female students (placed on top of male)
plt.bar(
    departments,
    female_students,
    bottom=male_students,   # Stack on top of male values
    color='pink',
    label='Female Students'
)

# Title and axis labels
plt.title('Male and Female Students in Departments')
plt.xlabel('Departments')
plt.ylabel('Number of Students')

# Show legend to differentiate categories
plt.legend()
```

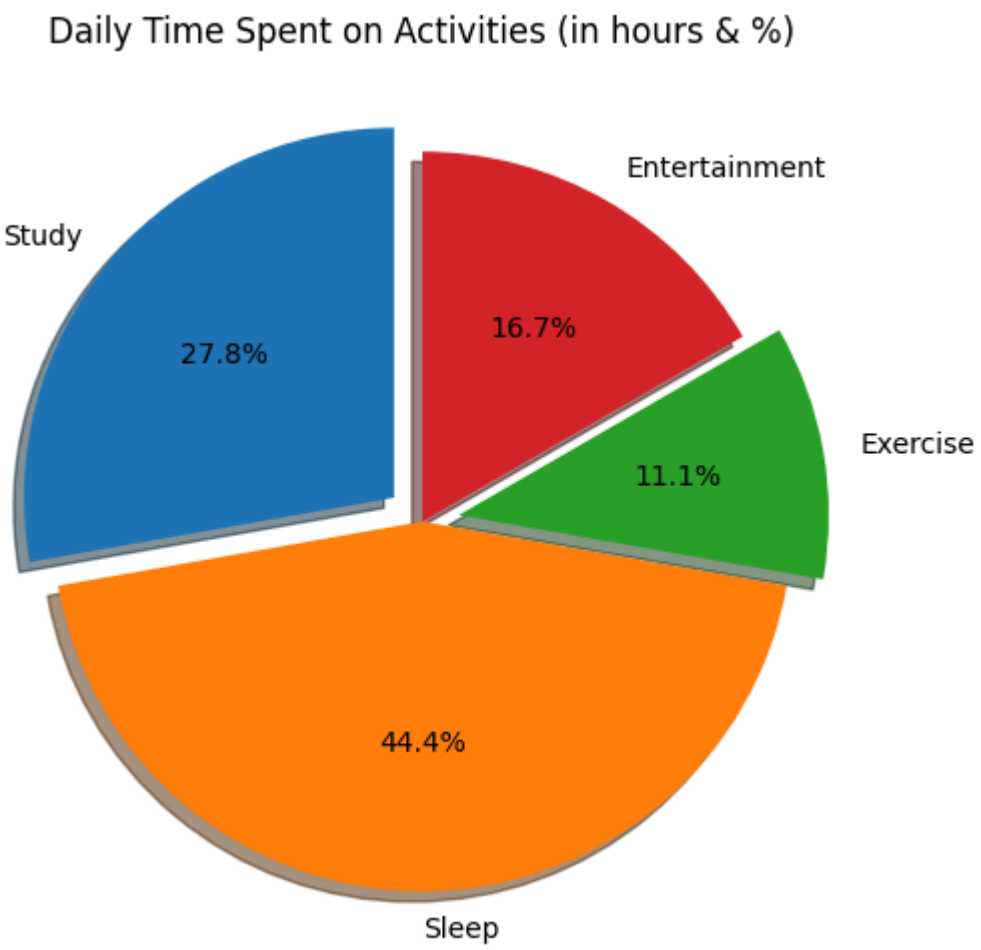
```
# Display the chart
plt.show()
```



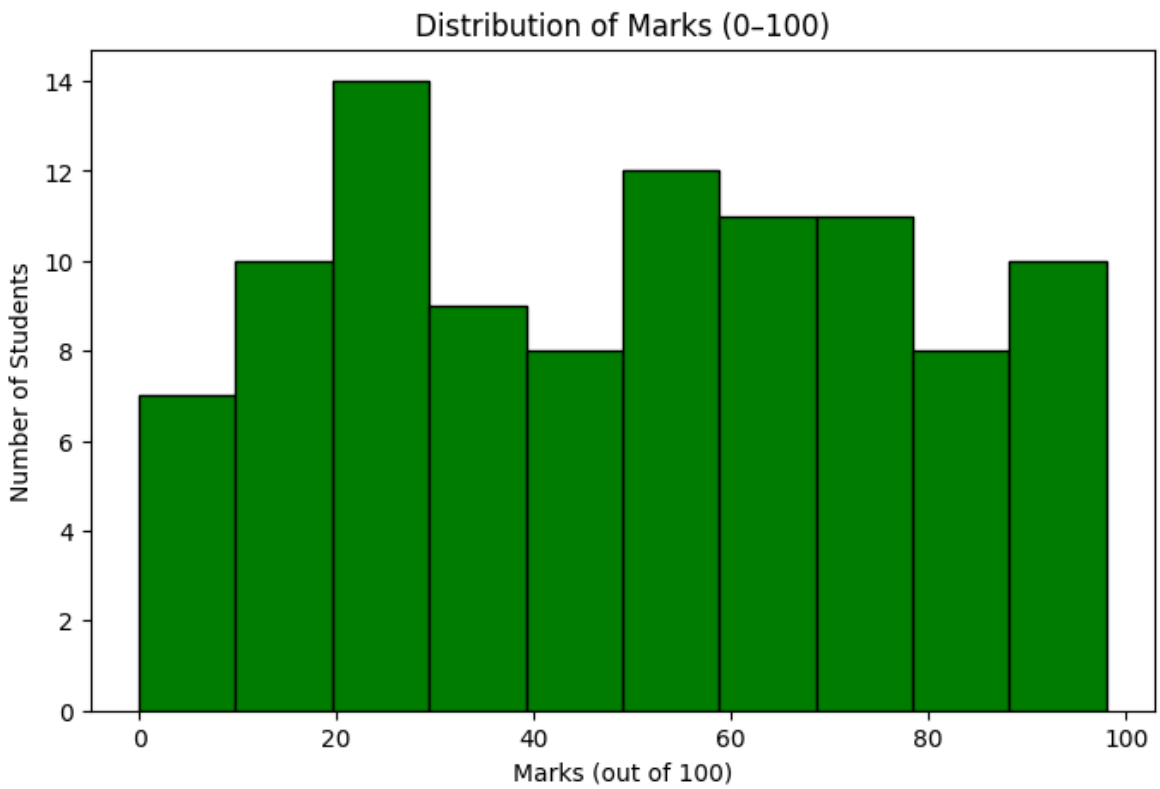
In [6]:
```
import matplotlib.pyplot as plt

# Smartphone brands
brands = ['Apple', 'Samsung', 'OnePlus', 'Xiaomi', 'Others']

# Market share data (in percentage %)
market_share = [30, 25, 15, 20, 10]

# Create figure
plt.figure(figsize=(6, 6))

# Create pie chart
plt.pie(
    market_share,
    labels=brands,              # Brand names as labels
    autopct='%1.1f%%',          # Show percentage values
    startangle=140              # Rotate chart for better view
)

# Title with unit
plt.title('Smartphone Market Share (in %)')

# Display chart
plt.show()
```



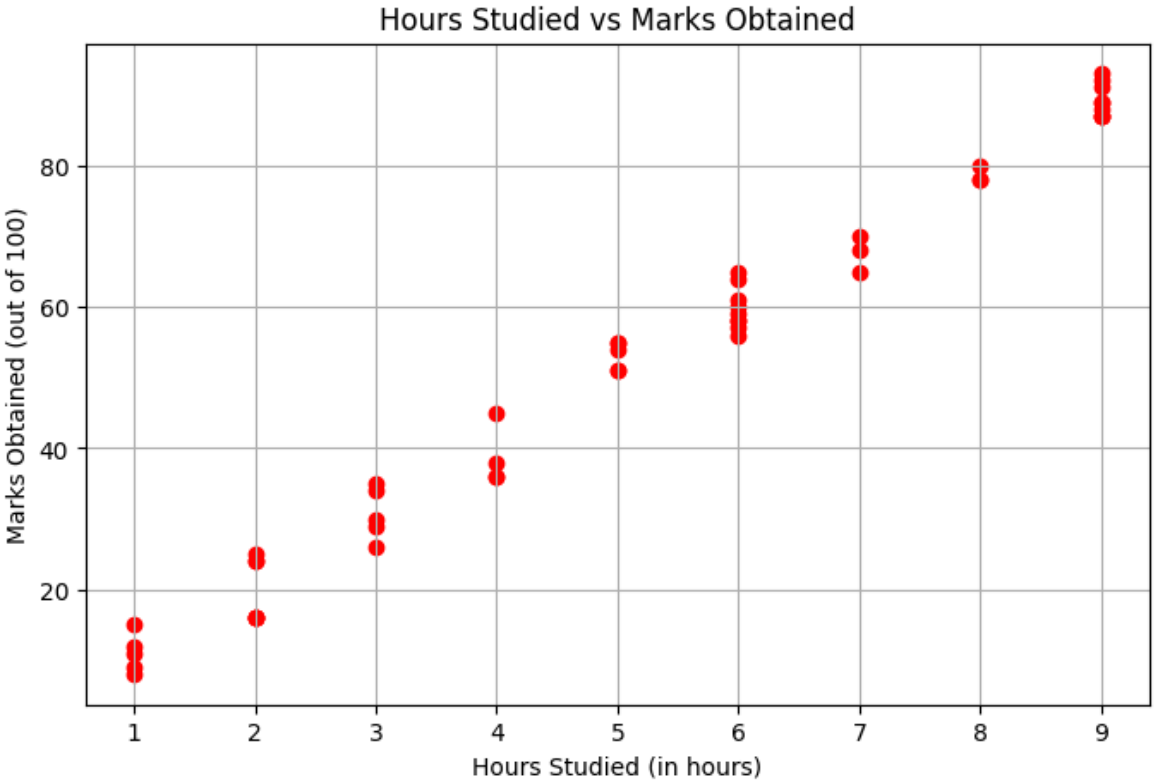
In [9]:
```
import matplotlib.pyplot as plt

# Daily activities
activities = ['Study', 'Sleep', 'Exercise', 'Entertainment']

# Time spent on each activity (in hours)
hours = [5, 8, 2, 3]

# Explode values to highlight specific slices (Study & Exercise)
explode = [0.1, 0, 0.1, 0]

# Create figure
plt.figure(figsize=(6, 6))

# Create pie chart
plt.pie(
    hours,
    labels=activities,          # Activity names
    autopct='%1.1f%%',          # Show percentage
    explode=explode,            # Highlight selected slices
    shadow=True,                # Add shadow effect
    startangle=90               # Start from top (90°)
)

# Title with unit context
plt.title('Daily Time Spent on Activities (in hours & %)')

# Display chart
plt.show()
```



In [10]:
```
import numpy as np
import matplotlib.pyplot as plt

# Generate marks for 100 students (range: 0 to 100)
marks = np.random.randint(0, 101, 100)

# Create figure
plt.figure(figsize=(8, 5))

# Create histogram
plt.hist(
    marks,
    bins=10,                # Number of intervals (bins)
    color='green',          # Bar color
    edgecolor='black'       # Border color of bars
)

# Title and axis labels
plt.title('Distribution of Marks (0-100)')
plt.xlabel('Marks (out of 100)')
plt.ylabel('Number of Students')

# Display the chart
plt.show()
```

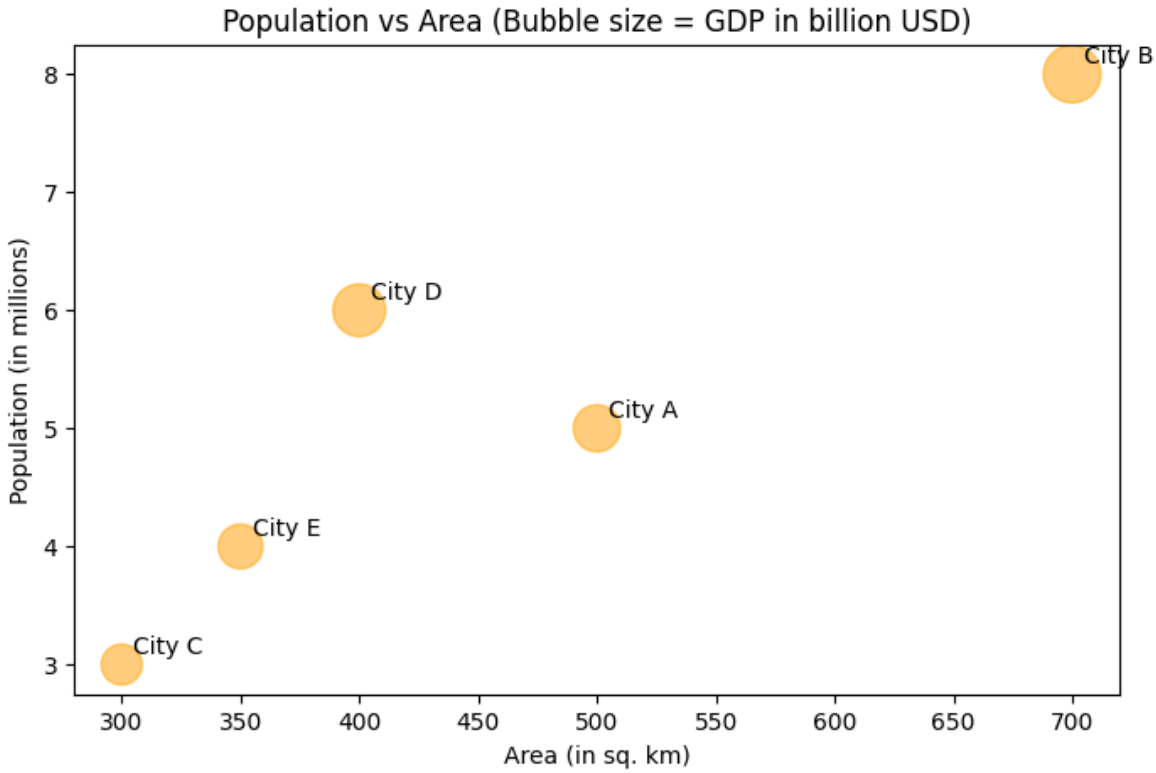


In [12]:
```
import numpy as np
import matplotlib.pyplot as plt

# Generate study hours (1 to 9 hours)
hours_studied = np.random.randint(1, 10, 50)

# Generate marks based on study hours (approx relation: marks ≈ hours * 10)
# Adding small random variation (-5 to +5) for realism
marks_obtained = hours_studied * 10 + np.random.randint(-5, 6, 50)

# Create figure
plt.figure(figsize=(8, 5))

# Create scatter plot
plt.scatter(
    hours_studied,
    marks_obtained,
    color='red',      # Point color
    marker='o'        # Circular markers
)

# Title and axis labels with units
plt.title('Hours Studied vs Marks Obtained')
plt.xlabel('Hours Studied (in hours)')
plt.ylabel('Marks Obtained (out of 100)')

# Add grid for better readability
plt.grid(True)

# Display the chart
plt.show()
```



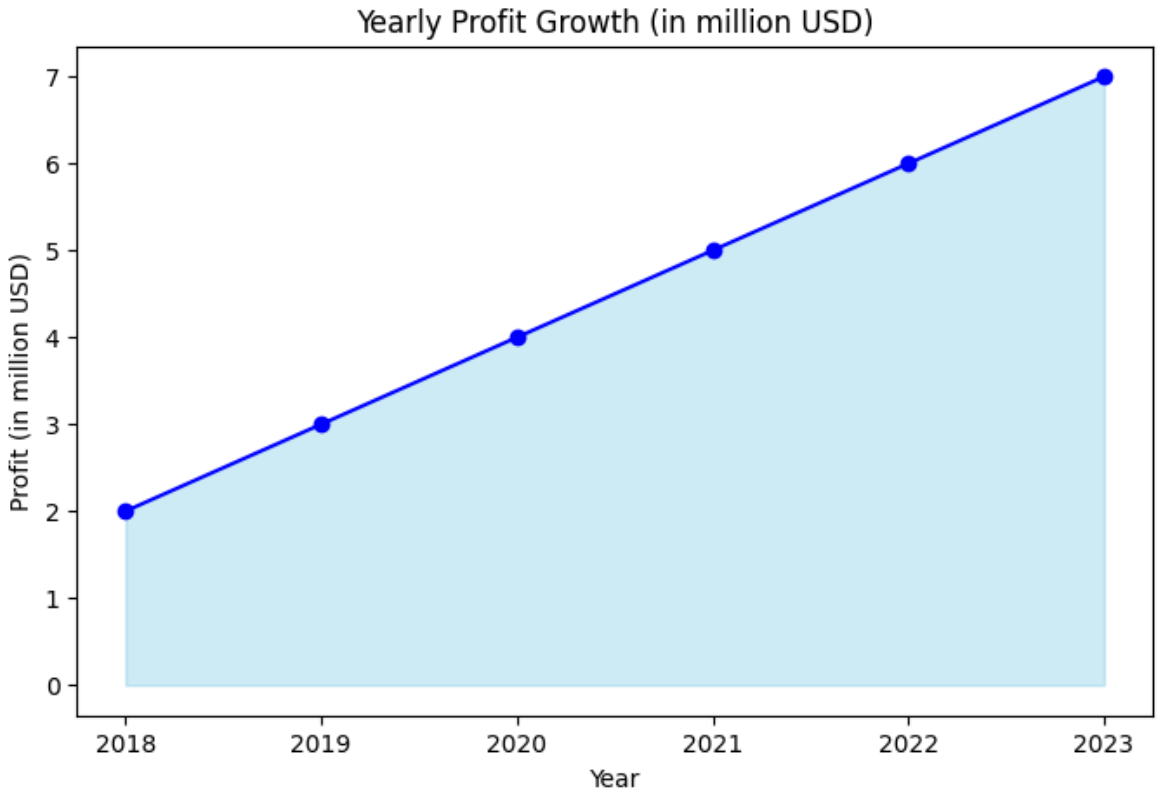
In [13]:
```
import numpy as np
import matplotlib.pyplot as plt

# City names
cities = ['City A', 'City B', 'City C', 'City D', 'City E']

# Data with units
population = [5, 8, 3, 6, 4]     # Population (in millions)
area = [500, 700, 300, 400, 350] # Area (in sq. km)
gdp = [200, 300, 150, 250, 180]  # GDP (in billion USD)

# Create figure
plt.figure(figsize=(8, 5))

# Create bubble chart
plt.scatter(
    area,
    population,
    s=np.array(gdp) * 2,   # Bubble size proportional to GDP
    alpha=0.5,             # Transparency
    color='orange'         # Bubble color
)

# Title and axis labels with units
plt.title('Population vs Area (Bubble size = GDP in billion USD)')
plt.xlabel('Area (in sq. km)')
plt.ylabel('Population (in millions)')

# Add city labels near each bubble
for i, city in enumerate(cities):
    plt.text(
        area[i] + 5,
        population[i] + 0.1,
        city
    )
```

```
# Display the chart
plt.show()
```

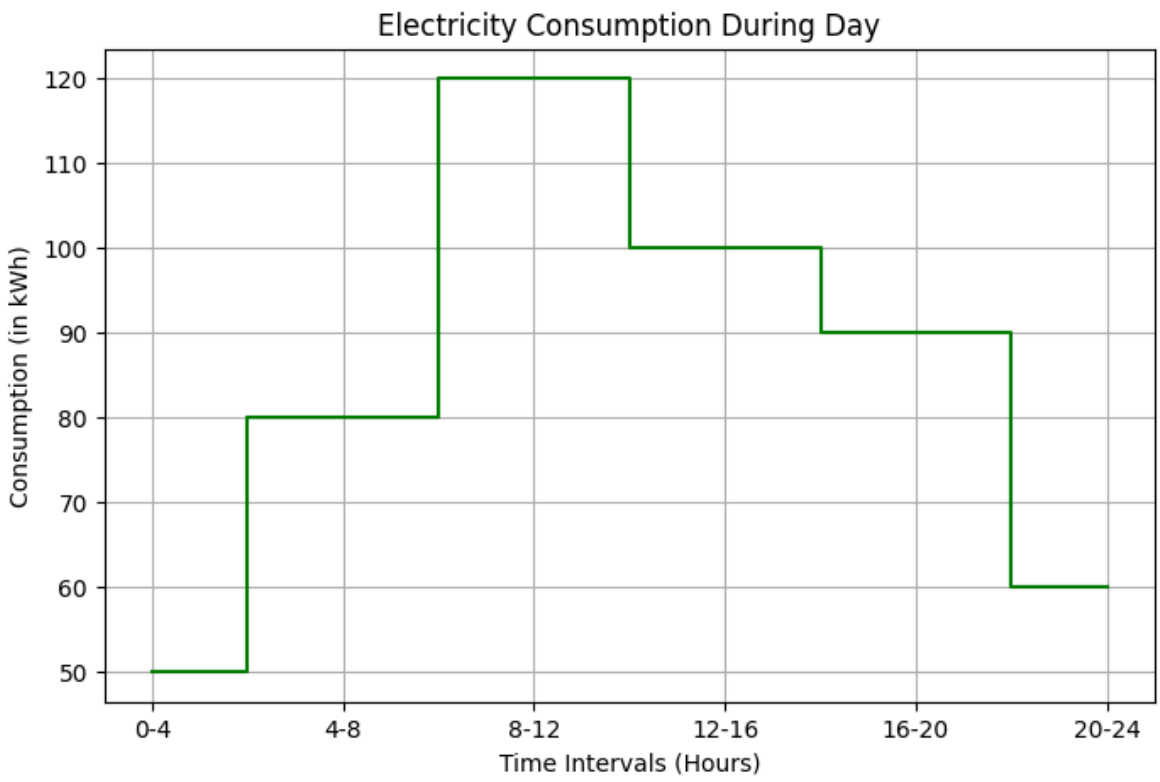


In [14]:
```
import matplotlib.pyplot as plt

# Years
years = [2018, 2019, 2020, 2021, 2022, 2023]

# Profit data (in million USD)
profit = [2, 3, 4, 5, 6, 7]

# Create figure
plt.figure(figsize=(8, 5))

# Create area chart (filled region)
plt.fill_between(
    years,
    profit,
    color='skyblue',    # Fill color
    alpha=0.4           # Transparency
)

# Plot line on top of area
plt.plot(
    years,
    profit,
    color='blue',
    marker='o'          # Mark each data point
)

# Title and axis labels with units
plt.title('Yearly Profit Growth (in million USD)')
plt.xlabel('Year')
plt.ylabel('Profit (in million USD)')
```

```
# Display the chart
plt.show()
```

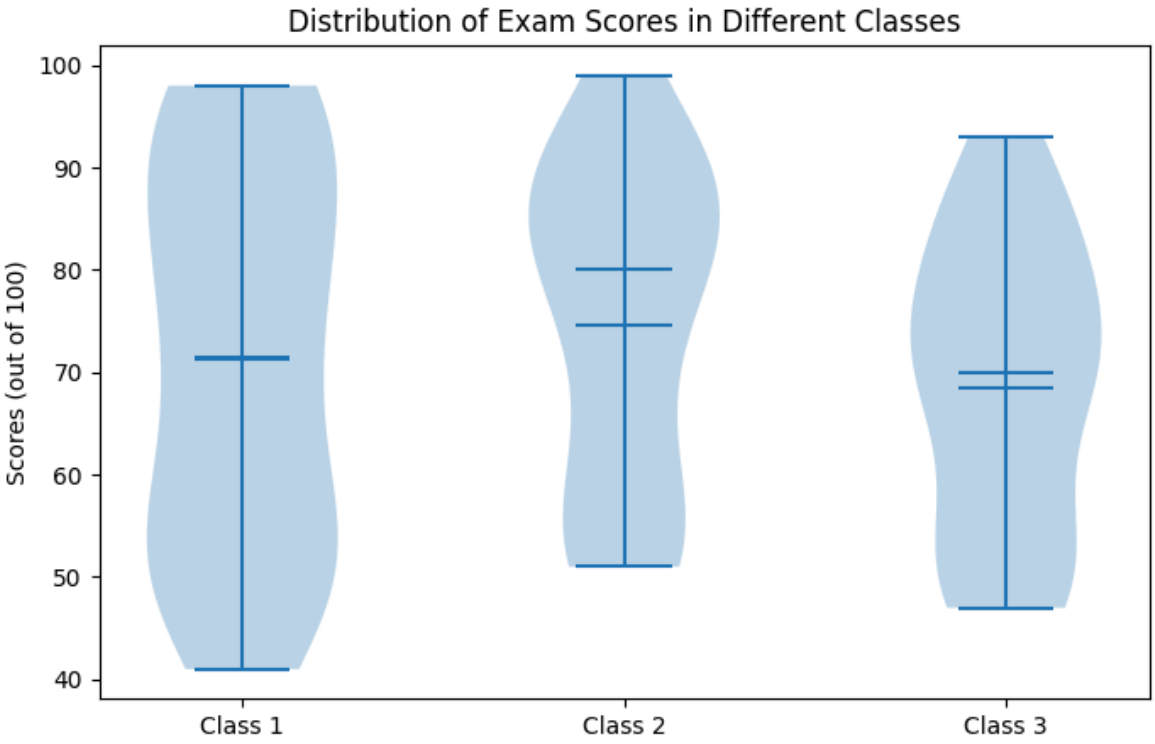


In [16]:
```
import matplotlib.pyplot as plt

# Time intervals of the day (in hours)
time_intervals = ['0-4', '4-8', '8-12', '12-16', '16-20', '20-24']

# Electricity consumption data (in kWh)
consumption = [50, 80, 120, 100, 90, 60]

# Create figure
plt.figure(figsize=(8, 5))

# Create step chart
plt.step(
    time_intervals,
    consumption,
    where='mid',      # Step changes at middle of interval
    color='green'     # Line color
)

# Title and axis labels with units
plt.title('Electricity Consumption During Day')
plt.xlabel('Time Intervals (Hours)')
plt.ylabel('Consumption (in kWh)')

# Add grid for better readability
plt.grid(True)

# Display the chart
plt.show()
```



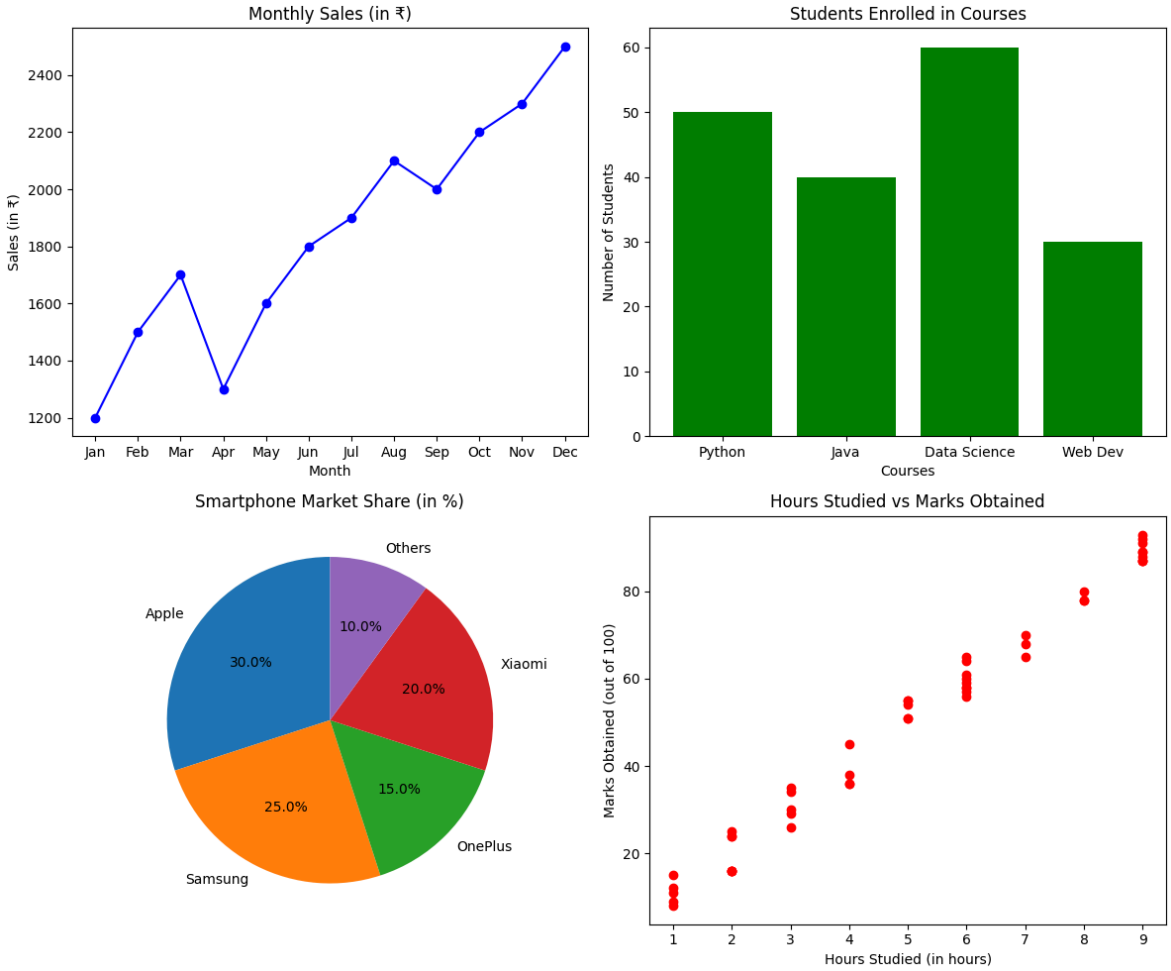
In [18]:
```
import numpy as np
import matplotlib.pyplot as plt

# Generate salary data for 3 departments (in Indian Rupees ₹)
dept1 = np.random.randint(30000, 70000, 20)
dept2 = np.random.randint(35000, 80000, 20)
dept3 = np.random.randint(40000, 90000, 20)

# Create figure
plt.figure(figsize=(8, 5))

# Create box plot (updated parameter name)
plt.boxplot(
    [dept1, dept2, dept3],
    tick_labels=['Dept 1', 'Dept 2', 'Dept 3']   # Updated from 'labels'
)

# Title and axis label with units
plt.title('Salary Distribution of Employees (in ₹)')
plt.ylabel('Salary (in ₹)')

# Display the chart
plt.show()
```

In [19]:
```
import matplotlib.pyplot as plt
import numpy as np

# Generate exam scores for 3 classes (range: 0–100 marks)
class1 = np.random.randint(40, 100, 30)
class2 = np.random.randint(50, 100, 30)
class3 = np.random.randint(45, 95, 30)

# Combine data
data = [class1, class2, class3]

# Create figure
plt.figure(figsize=(8, 5))

# Create violin plot
plt.violinplot(
    data,
    showmeans=True,     # Show mean value
    showmedians=True    # Show median value
)

# Set x-axis labels (modern & clear)
plt.xticks(
    [1, 2, 3],
    ['Class 1', 'Class 2', 'Class 3']
)

# Title and axis labels with units
plt.title('Distribution of Exam Scores in Different Classes')
plt.ylabel('Scores (out of 100)')

# Display the chart
plt.show()
```



In [20]:
```
import matplotlib.pyplot as plt

# Create overall figure
plt.figure(figsize=(12, 10))

# Line Chart: Monthly Sales (in ₹)
plt.subplot(2, 2, 1)
plt.plot(
    months,
    monthly_sales,
    marker='o',
    color='blue'
)
plt.title('Monthly Sales (in ₹)')
plt.xlabel('Month')
plt.ylabel('Sales (in ₹)')

# Bar Chart: Students Enrolled
plt.subplot(2, 2, 2)
plt.bar(
    courses,
    students_enrolled,
    color='green'
)
plt.title('Students Enrolled in Courses')
plt.xlabel('Courses')
plt.ylabel('Number of Students')

# Pie Chart: Market Share (in %)
plt.subplot(2, 2, 3)
plt.pie(
    market_share,
    labels=brands,
    autopct='%1.1f%%',
    startangle=90
)
```

```
plt.title('Smartphone Market Share (in %)')

# Scatter Plot: Study vs Marks
plt.subplot(2, 2, 4)
plt.scatter(
    hours_studied,
    marks_obtained,
    color='red'
)
plt.title('Hours Studied vs Marks Obtained')
plt.xlabel('Hours Studied (in hours)')
plt.ylabel('Marks Obtained (out of 100)')

# Adjust layout to prevent overlap
plt.tight_layout()

# Display all plots
plt.show()
```



In [ ]: